The Direct and Indirect Effects of Genetics and Education

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Abstract

Social science genetics has shown genes matter for education and labour market outcomes, but has little credible evidence on the economic mechanisms beneath these findings. I use plausibly random deviation from parents as a natural experiment for the Education PolyGenic Index (Ed PGI) to estimate the causal effect of genes associated with years of education, with data from the UK Biobank. I then decompose the total genetic genetic effects into a direct genetic effect and indirect education pathway using a causal mediation framework that allows for selection-into-education, using multiple UK university openings in the 1960s as instruments for higher education attendance. The Ed PGI increases labour market earnings through higher education attendance, with direct genetic effects indistinguishable from zero — contradicting previous speculation that education-linked genes independently increase intelligence or other labour market traits. Education linked genes are more about returns to higher education, and less about direct genetic effects. These findings reframe the usual focus in social science genetics by shifting from asking "if genes matter" to "how genes matter" — not as deterministic forces, but as factors that influence human capital decisions within institutional constraints. By using causal methods that account for individual choice in education, and not than assuming genetic determinism, this research provides a richer understanding of how biological inheritance, human capital decisions, and institutional structures interact to shape labour market inequality.

Keywords: Education, Mediation, Polygenic indicies, Nature/nurture.
JEL Codes: D31, D91, I24, J24, Z00.

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> The difference of natural talents in different men, is, in reality, much less than we are aware of [...]. The difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise not so much from nature, as from habit, custom, and education.
>
> — *Smith (1776), The Wealth of Nations, Volume 1 p. 16.*

0.1 Writing Plan

The introduction formula https://blogs.ubc.ca/khead/research/research-advice/formula.

Section map for this paper:

1. Intro, written last following https://blogs.ubc.ca/khead/research/research-advice/formula.
2. Data + biological context.
3. Section on the adjusted PGI randomisation design, clearly stating estimands
4. Long section of results. Add a subsection of a discussion of the results: note that mechanisms are unclear. Without additional random variation in the education channel, it could entirely be the case that genetic effects are returns to education, and genetic effects pick up institutional inaccess and inequalities in the British education system for people born 1940 to 1970 (of which there were obviously many). One small figure showing the correlation between Ed PGI and other PGIs, that it is negatively correlated with other indicies; and a similar one for the random component.
5. Conclusion.

Note: a major concern for the Ed PGI previous papers is that they have little economic insight. This paper is framed as changing that, investigating the economic mechanisms underlying the Ed PGI causal effects, giving economic credence where before there was merely concern.

Papageorge & Thom (2020) document how the Ed PGI predicts earnings, even when controlling for education, and posit that the index contains genes that affect skills related to financial literacy.

Quote for the Mendelian independent first-stage, "Cannot choose who your parents are, and you also cannot choose how much you inherit from them."

1 Introduction

This paper proceeds as follows. Section 2 summarises the UKB data, and explains how the Ed PGI is constructed. Section 3 shows a causal research design, based on Mendelian independent inheritance, for estimating the causal effects of the Ed PGI; Section 4 shows estimates of the causal effects of the Ed PGI, on education and labour market outcomes. Section 5 gives a conceptual framework for estimating the direct and indirect effects of genetics and education, in this observational setting; Section 6 gives the empirical results. Section 7 concludes.

2 Data and Genetic Context

2.1 UK Biobank (UKB)

The UK Biobank (UKB) is a large-scale prospective cohort study comprising approximately half a million members of the British public, recruited between 2006 and 2010. Letters were sent to a random sample of addresses in the National Health Service registry, and the final UKB data represent voluntary respondents.[1] Participants, aged 40-69 years at recruitment, underwent comprehensive baseline assessments, asked many questions for socioeconomic information either by interview, automated survey answers, or follow-up sessions. Key socioeconomic variables include educational attainment (measured in education qualifications), household income brackets, employment status, and industry of employment. See Sudlow et al. (2015) for a full description of the UKB organisation and data.

The UKB collected detailed information on educational qualifications, including completion of primary school, GCSEs (typically completed at age 16), A-Levels (completed at age 18), and completion of an higher education degree. I convert these qualifications into education years based on each participant's highest reported qualification, following the International Standard Classification of Education.[2] While the UKB collected household income data, this information was recorded only in categorical bins, limiting its utility for continuous

[1] The UKB may not be representative of the entire UK population, as it only represents willing respondents. As such, the findings of this paper should only be taken as internally valid among the UKB respondents, and the sibling analysis sample described.

[2] As noted by Okbay et al. (2022), British vocational degrees inconsistently coded by the International Standard Classification of Education. I follow the literature by assigning education years as the age they reported leaving full-time education minus five, among those with a vocational degree as their highest qualification.

Table 1: Descriptive Statistics, UKB.

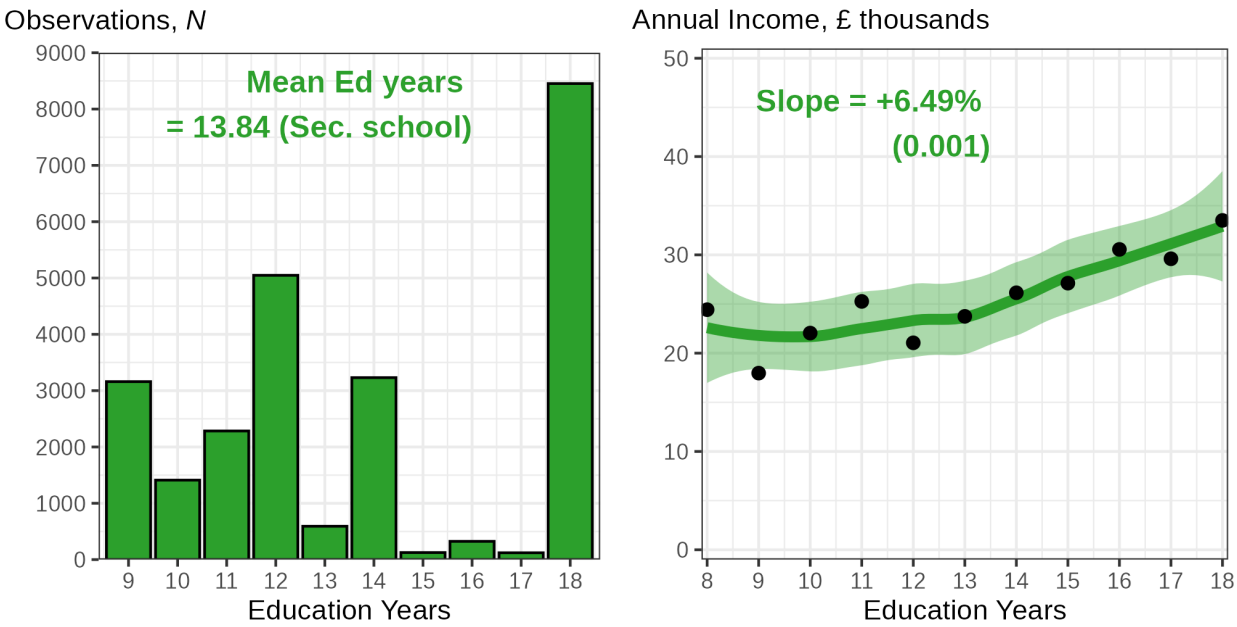
	Analysis sample		Entire sample	
	Mean	SD	Mean	SD
Demographics:				
Male	0.44	0.50	0.46	0.50
Age	55.04	7.05	56.55	8.09
Lives in city	0.86	0.35	0.85	0.35
Ethnicity = European	1.00	0.00	0.82	0.38
Any siblings also in UKB?	1.00	0.00	0.08	0.28
Count of siblings in UKB	1.09	0.32	0.09	0.32
Genetic Measures:				
Ed PGI	0.04	1.00	-0.00	1.00
Ed PGI, imputed parental mean	-0.02	0.75		
Other PGI: ADHD	-0.02	1.00	-0.00	1.00
Other PGI: Asthma	-0.01	1.00	0.00	1.00
Other PGI: Bipolar	-0.01	1.00	-0.00	1.00
Other PGI: BMI	-0.02	0.99	-0.00	1.00
Other PGI: Diabetes (T2)	-0.02	1.00	-0.00	1.00
Other PGI: Height	0.00	1.01	-0.00	1.00
Other PGI: Schizophrenia	-0.04	1.00	0.00	1.00
Education:				
Education years	13.84	3.36	13.72	3.48
Age left education	16.67	2.47	16.51	3.04
Qualification, University degree	0.33	0.47	0.33	0.47
Qualification, A-Levels	0.46	0.50	0.45	0.50
Qualification, GCSEs	0.76	0.43	0.71	0.45
Qualification, Professional degree	0.32	0.47	0.29	0.45
Qualification, Vocational degree	0.23	0.42	0.19	0.39
Qualification, No official qualifications	0.13	0.34	0.17	0.38
Labour Market Outcomes:				
Occupation hourly wage, £	18.63	8.00	19.23	8.31
Occupation annual income, thousands £	26.33	16.57	27.83	17.48
Average hours worked, per week	34.73	12.58	35.31	12.70
Household income, $< £18k$	0.13	0.33	0.20	0.40
Household income, $£18 - 31k$	0.23	0.42	0.22	0.41
Household income, $£31 - 52k$	0.28	0.45	0.22	0.42
Household income, $£52 - 100k$	0.22	0.41	0.17	0.38
Household income, $£100k <$	0.05	0.22	0.05	0.21
Observations	24,755		496,204	

Note: This table shows the Mean and Standard Deviation (SD) of UKB variables, comparing the analysis subsample and the entire UKB sample. The analysis sample refers to individuals with at least one sibling in the UKB data, of European ancestry. All genetic variables are only non-missing among those who report European ancestry.

analysis. Following Kweon et al. (2025), I construct an individual-level income measure using participants' 4-digit Standard Occupational Classification (SOC) codes. Specifically, I impute occupation-based wages by estimating hourly earnings for each SOC category using data from the British Annual Survey of Hours and Earnings, conditional on age, gender, and year. This approach provides a more granular and continuous measure of economic status than the categorical household income variable (see Appendix A.1 for details on the imputation procedure). I create the annual occupation income by multiplying this by the number of hours they reported working, times 52 weeks in the year. For example, this procedure gives a female participant, age 58 in 2010 with SOC 2314 (i.e., a secondary education teacher by profession), an occupation hourly wage of £29.20 and an annual income of £68,476 for her 45 hours worked each week (adjusted to 2024 £GBP by CPI-H).

Figure 1: Distribution of UKB Education Measures.

(a) Education Years. **(b)** Education Years + Income.



Note: This figure shows the marginal distribution of education years in the UKB analysis sample, and the correlation between years of education and occupation coded income.

For the purposes of this study, the primary analysis sample is restricted to those with at least one family member, sibling or parent, also enrolled in the UKB, and non-missing data in occupation code plus education qualifications. The UKB does not personally identify participants, nor their family members, so I follow the genetic literature by identifying family clusters by measures of genetic similarity.[3] The presence of multiple family members within

[3]The design of the UKB allows for a larger number of family clusters than would be expected from a

the dataset provides a unique opportunity to control for shared genetic factors that might otherwise bias estimates in standard cross-sectional analyses.

The analysis sample is also restricted to individuals who report to have primarily European ancestry. The genetic measures used in this paper have generally been constructing from primarily white, European descended populations, and have no predictive power out of sample, so are only relevant in this subsample.[4] The only notable differences between the analysis subsample and the entire UKB participant sample is that the analysis sample has more siblings, and is entirely European descended; no other variables are significantly different for the analysis subsample. Table 1 summarises the UKB data, for the analysis sample and among the entire UKB sample. Figure 1 summarises the correlation between education years and occupation coded income.

2.2 Education PolyGenic Index (Ed PGI)

UKB participants also provided saliva samples that were subsequently genotyped using high-density arrays containing over 800,000 genetic markers, producing standardised genetic data files for statistical analysis of the relationship between genetics and social outcomes.

While genetic data provide rich information about individual variation, the relationship between specific genes and social outcomes like education is extraordinarily complex. The biological pathways from DNA to observable traits remain largely opaque — the exact relationship between a single gene, its resulting protein (or its moderating role), and the effects of those proteins are only known for a small number of genes in the human genome. For example, a single gene (and its recessive variant) is known to lead to a protein which does not break down the by-products of alcohol digestion; individuals with this gene generally do not drink alcohol at all because of the simple limit on digestion (Millwood et al. 2023). Complex social behaviours, on the other hand, have no single genes with a known biological pathway, so cannot be summarised in a single gene, nor single biological pathway. Instead complex social behaviours are a result on thousands of genes, and their interactions with the environment — they are polygenic.

random sample of the British population. This is because study enrolment letters were sent to households, and the household head (or to whomever the letter was addressed) could optionally enrol their family members in addition to themselves. Appendix A.2 describes the family cluster identification procedure in detail, which has been used in previous social science genetic studies (see e.g., Muslimova et al. 2025).

[4]It is an unfortunate fact that systematic genetic measures, such as those considered in this paper, have only been constructed from Biobank samples where the vast majority are white and/or European descended; see Martin et al. (2017) for a further discussion. Note that this implies this paper has nothing to say about race; see Subsection 2.4.

Rather than attempting to trace all of these intricate biological mechanisms, researchers have developed PolyGenic Indices (PGIs) as dimensional reduction tools that summarise the statistical relationship between thousands of genetic variants and outcomes of interest.[5] This aggregation approach is particularly valuable because individual genetic variants typically have tiny effects on complex traits like educational attainment — often explaining less than 0.01% of the variation. However, by combining information across hundreds of thousands of variants, each contributing a small effect, PGIs can capture meaningful portions of the genetic contribution to these outcomes. These indices are fundamentally correlational measures: they capture statistical associations between genetic variants and outcomes observed in large samples, not well understood biological pathways, and without a clear case for causal relationships. The weights used to construct PGIs reflect population-level correlations that may arise through various channels, including biological effects, gene-environment correlations, and population stratification.

The Education PGI (Ed PGI) is a summary measure that aggregates genetic information across thousands of genetic variants into a single index predictive of educational attainment. The construction of the Ed PGI relies on weights derived from a Genome-Wide Association Study (GWAS), which analyse the correlation between genetic variants and education years. In a GWAS, researchers test the correlation between each Single Nucleotide Polymorphism (SNP) — a one-letter change in the DNA sequence at a specific location — and the outcome of interest, in this case education years. I use the weights calculated in the GWAS conducted by Okbay et al. (2022), which used a sample of over 3 million individuals of European ancestry, primarily from 23&Me data.

The Ed PGI is defined as the weighted sum of whether individual i has the relevant SNP at each DNA location j,

$$Z_i = \sum_{j=1}^{J} w_j x_{i,j}$$

where w_j are the Ed PGI weights (provided in the supplementary data to Okbay et al. 2022), and $x_{i,j}$ is an indicator of whether individual i has the relevant SNP at DNA location j.[6] The Okbay et al. (2022) GWAS found that over 8,000 individual SNPs were significant in predicting education years, after passing a battery of biostatistical tests and finding no

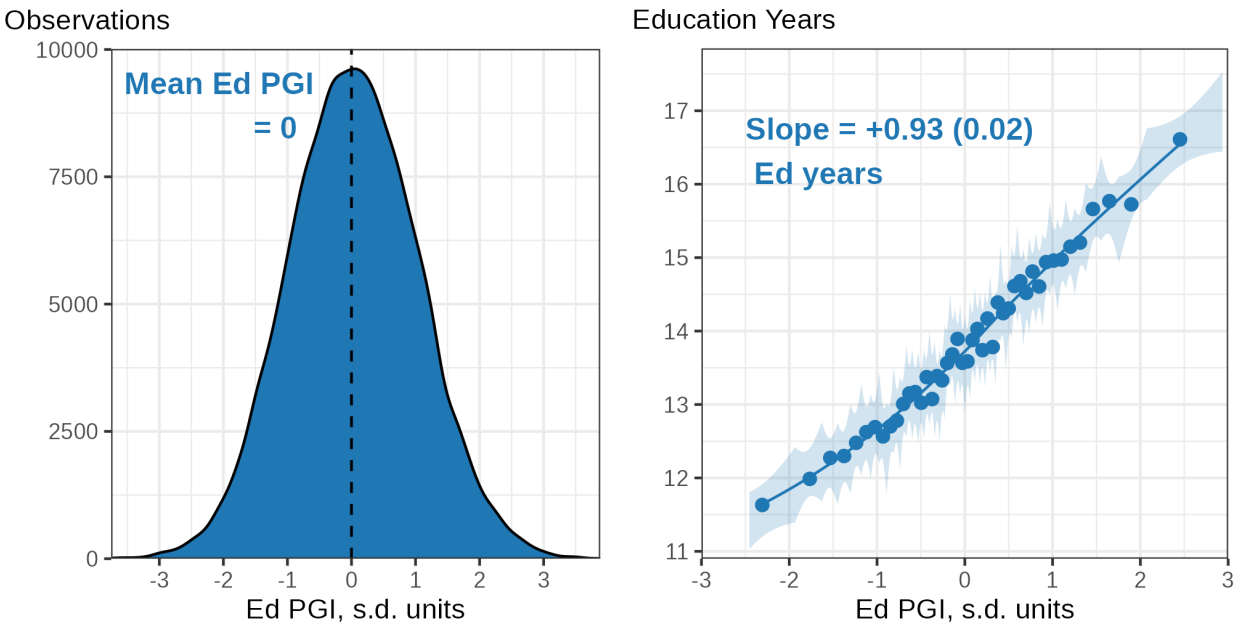
[5] See Uffelmann et al. (2021) for a full explanation of how a GWAS works, and the approach to calculating PGIs.

[6] Every person has two copies of each chromosome, so that $x_{i,j}$ takes values 0, 1, 2 corresponding to how many copies of the SNP i has. The genetics literature has generally accepted that complex social behaviours are explained by additive effects (see Hill et al. 2008), and Okbay et al. (2022) found little evidence for dominance in the Ed PGI SNPs, so that the Ed PGI makes sense as a weighted sum.

evidence for significant interactions. The resulting Ed PGI is standardised to have mean zero and standard deviation one within the entire sample, making it interpretable as the number of standard deviations from the population mean. The Ed PGI captures approximately 12-15% of the variation in educational attainment, making it one of the most predictive PGIs for any complex behavioural trait to date.

Figure 2: Distribution of Education PolyGenic Index (Ed PGI).

(a) Ed PGI. **(b)** Ed PGI + Education Years, Binscatter.



Note: This figure shows the marginal distribution of Ed PGI and education years in the UKB analysis sample. Ed PGI is normalised to have mean 0 and standard deviation 1 among the entire UKB sample; it is roughly normally distributed regardless of this normalisation.

The Ed PGI is highly correlated with education years, as it is constructed to be so. A one standard deviation increase in the Ed PGI is associated with 0.93 extra education years (standard error 0.02), with an R^2 value of 0.10 for the raw regression of education years on Ed PGI. It is roughly normally distributed, with standard deviation normalised to 1. It is not surprising that the Ed PGI is normally distributed; it is the weighted sum of thousands of binary variables, among a few thousand individuals, so we should expect the central limit theorem to apply. Additionally, it is not surprising that the Ed PGI is correlated with education years; it was literally constructed to be so. The main take-away is that the correlation is strong, and has meaningful predictive power for education years. If education had no relationship with genetic variants, then we may expect the correlation would be indistinguishable from zero.

The Ed PGI is not the only PGI that can be constructed. Following the same methodology, I construct additional PGIs for six health-related traits that may confound the relationship between genetics and socioeconomic outcomes: asthma, bipolar disorder, Body Mass Index (BMI), height, schizophrenia, and type 2 diabetes. Each PGI is calculated using the same approach as the Ed PGI — taking weights from a published GWAS, conducted on independent populations, and computing weighted sums of the relevant genetic variants in the UKB data. These other PGIs serve as control variables in my analysis for two reasons. First, genetic variants often exhibit pleiotropy, meaning a single variant can influence multiple traits simultaneously. For instance, genetic variants associated with education might also affect other factors (e.g., mental health conditions), which could independently influence labour market outcomes. Second, systematic differences in gene frequencies across ancestry groups can create spurious correlations between any PGI and outcomes if health conditions vary across these groups for unrelated reasons.

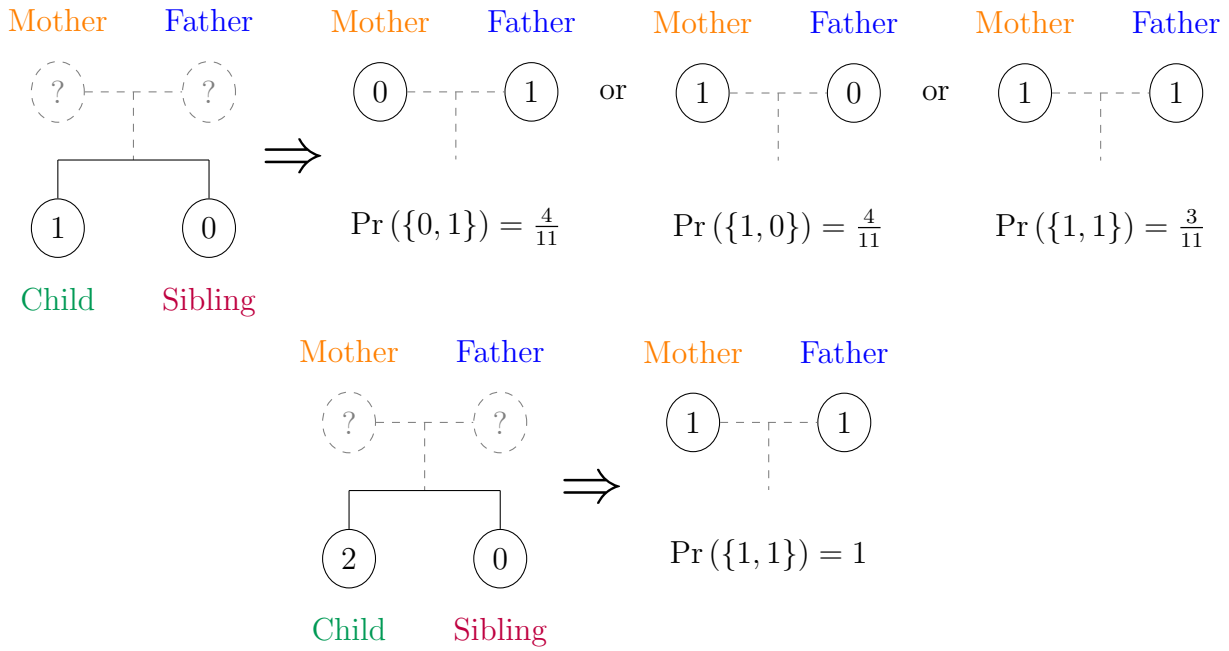
2.3 Imputing Parental PGIs from Siblings

It would be ideal if the UKB contained genetic information for the parents of every UKB participant. While there are 6,239 participants with a parent also included in the UKB, there are only 72 with both parents. To overcome this missing data problem, I use siblings' genetic data to impute the mean parental PGIs from the sample of 24,755 participants with at least one sibling also in the UKB — the sibling analysis sample described above.

I use the Young et al. (2022) procedure, to impute mean parental PGI values using genetic data on participants' siblings. To briefly summarise how this works, consider that each person inherits exactly half their DNA from their mother and half from their father. When we observe two siblings, we can partially reconstruct their parents' genetic information because siblings both inherit their DNA from parents. Consider observing two different samples from the same underlying distribution — the parents' genetics. At some genetic locations, siblings inherited the same variant from a parent (like both getting mum's brown-eye variant), whilst at others they inherited different variants (one got mum's brown-eye variant, the other her blue-eye variant). By comparing siblings' genetics across thousands of genetic variants, we can recover about 75% of their parents' combined genetic information — enough to construct a reasonably accurate parental PGI.

Figure 3 visualises the logic behind imputing parental SNP values based on an individual's, and their sibling(s)'s, genetic data. In the middle panel, the individual has genotype 1 and their sibling has genotype 0. Since the individual must have inherited their 1 variant from

Figure 3: Imputing One SNP for Parents from Sibling Data.



Note: This figure illustrates the logic of imputing parental genotypes from sibling data at a single SNP. At each SNP, individuals have a genotype of 0, 1, or 2, representing the number of copies of a particular variant inherited from their parents. A child, and their sibling, jointly imply the possible options for the parents. The first option comes from the fact that child and sibling configuration of $\{1, 0\}$ is only possible with the configuration of parents of $\{0, 1\}, \{1, 0\}$ or $\{1, 1\}$. With a parent configuration of $\{0, 1\}$ or $\{1, 0\}$, there are 4 possible realisations with a child and a sibling taking $\{1, 0\}$, and 3 for parental configuration of $\{1, 1\}$. The second option is only possible with one parent configuration, so their is no uncertainty in what the parent values were. This gives the relevant probabilities as a result of this combinatorial mixing, which is also shown in Table 2.

one parent and their 0 variant from the other, while the sibling inherited two 0 variants, we can infer that one parent has genotype 0 (contributing 0 to both children) and the other has genotype 1 (contributing 1 to the individual and 0 to the sibling). The right panel shows an alternative configuration yielding the same sibling genotypes. When parental genotypes cannot be uniquely determined, the imputation uses the expected values based on observed variants and population frequencies. By applying this logic across hundreds of thousands of SNPs and accounting for segments of DNA shared identically by descent between siblings, we can accurately reconstruct parental PGIs despite never observing the parents directly.

The imputation procedure performs well in the UKB analysis sample. Following the Young et al. (2022) validation approach, I verify the accuracy by examining the small subset of families where parental genotypes are actually observed. The imputed parental PGIs show a correlation of over 99% with true parental PGIs, and regression coefficients are statistically

indistinguishable from 1, confirming the imputation is unbiased. Appendix A.3 describes the full technical detail behind the imputation procedure, and this validation exercise. This high accuracy means I can confidently use the imputed parental PGIs in place of directly observed parental PGI values.

2.4 Ethical Considerations

It is crucial to emphasise that the genetic measures in this study should not be misinterpreted as evidence of genetic determinism. Having a higher Ed PGI does not destine someone to higher education or income; it merely represents only a small statistical tendency, not fate. These tendencies are fundamentally shaped by social policies, institutional structures, and cultural contexts that vary across time and place. Just as eyesight correcting glasses can eliminate the practical consequences of genetic variation in eyesight, education and economic policies can modify or eliminate the consequences of genetic variation in traits relevant to socioeconomic outcomes.

Furthermore, this research provides no support whatsoever for ideas about "natural" social hierarchies or genetic justifications for inequality. In addition, genetic associations identified within one ancestry group cannot be meaningfully compared across groups due to different genetic architectures, environmental contexts, and historical experiences. Any claim using the measures consider in this paper for inference across racial groups, or different racial groups would be both scientifically invalid, and ethically deficient. The appropriate interpretation of these results is that they highlight how both social and genetic factors influence socioeconomic outcomes, which should inspire greater solidarity and support for policies that promote equality of opportunity rather than acceptance of inequality as natural or inevitable.

3 Random Variation in Genetic Inheritance

Genetic variants are not randomly assigned, which means that simple correlations between the Ed PGI and observed outcomes cannot straightforwardly be interpreted as causal. Genetic variation arises from inheritance: at conception, individuals inherit 23 chromosomes from each parent, obtaining half of their genetic variants from their mother and half from their father. Because parents transmit genetic material to their children, those who inherit a higher Ed PGI typically also inherit familial environments shaped by parents who themselves

possess a higher Ed PGI. Consequently, correlations between an individual's Ed PGI and outcomes may reflect not only direct genetic effects but also indirect effects mediated through family environment, making these correlations confounded.

However, while genetic inheritance generates systematic associations between genes and environments, it also includes a random component. Specifically, Mendelian inheritance ensures that, conditional upon parental genetics, there is random variation in the genetic variants a child inherits. This provides an Instrumental Variable (IV) to isolate causal effects of Ed PGI on outcomes. I exploit such Mendelian random variation to instrument for changes in the Ed PGI, thereby identifying causal relationships independent of confounding environmental factors.

3.1 Mendelian Independent Inheritance

I use variation away from parental values to estimate causal effects of having a higher versus lower Ed PGI. This approach leverages the fundamental randomness in genetic inheritance, first described by Mendel's laws (Mendel 1865).

Each parent has 0, 1, or 2 copies of each genetic variant — corresponding to presence on each of their chromosomes. When a father and mother's sex cells are formed, exactly one of the two copies is sorted into each reproductive cell, and is then inherited by their resulting child. A parent with one copy therefore passes it on half the time, whereas a parent with 0 or 2 copies transmits their genetic variant with certainty. At conception, the child receives one draw from the mother and one from the father, so the child's copy-count is simply the sum of two independent draws.

Write $x_{i,j}^{\text{Child}}$ for the number of copies of variant j that individual i has (0, 1, or 2), and $x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}$ for their respective parents $m(i), f(i)$. If the mother has two copies of variant j then they bequeath one copy to their child with certainty; if the father has one, then they bequeath one copy with probability $\frac{1}{2}$; if no copies, then they bequeath none with certainty. The child then receives their other copy from their father, by the same process, giving the following representation in expectation,

$$\mathbb{E}\left[x_{i,j}^{\text{Child}} \,\middle|\, x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}\right] = \frac{1}{2}\left(x_{m(i),j}^{\text{Mother}} + x_{f(i),j}^{\text{Father}}\right).$$

After conception, the child has inherited genetic variants from both parents — one of the 4 possible discrete realisations, not the expected value. This result came about purely from

Table 2: Mendelian Independent Inheritance of One Genetic Variant.

(a) Possible $x_{i,j}^{\text{Child}}$, given parental values.

		$x_{f(i),j}^{\text{Father}}$ 0	1	2
$x_{m(i),j}^{\text{Mother}}$	0	0,0,0,0	0,1,0,1	1,1,1,1
	1	0,0,1,1	0,1,1,2	1,1,2,2
	2	1,1,1,1	1,2,1,2	2,2,2,2

(b) Expected $x_{i,j}^{\text{Child}}$, given parental values.

		$x_{f(i),j}^{\text{Father}}$ 0	1	2
$x_{m(i),j}^{\text{Mother}}$	0	0	0.5	1
	1	0.5	1	1.5
	2	1	1.5	2

Note: This figure shows the possible inherited values of one SNP, given parental values. The 4 values in the table (a) shows the 4 possible realisations that parents with variants $x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}$. The values in table (b) show the expected value — a simple mean of the 4 realisations in table (a).

random chance, conditional on at least one of parent having exactly one copy of the SNP, $x_{m(i),j}^{\text{Mother}} = 1$ and/or $x_{f(i),j}^{\text{Father}} = 1$.[7]

$$x_{i,j}^{\text{Random}} := x_{i,j}^{\text{Child}} - \mathbb{E}\left[x_{i,j}^{\text{Child}} \,\middle|\, x_{m(i),j}^{\text{Mother}}, x_{f(i),j}^{\text{Father}}\right] \longrightarrow \text{Independently assigned.} \tag{1}$$

$x_{i,j}^{\text{Random}}$ is the random variation in presence of SNP at position j. From the possible values of the parents' variants, $x_{i,j}^{\text{Random}}$ equals zero if both parents have 0 or 2 copies, equals $\frac{1}{2}$ if the child actually inherited 1 copy when both parents had one copy each ($-\frac{1}{2}$ if they did not inherit), and 1 if they inherited 2 copies and the parents had one each (-1 if they inherited 0).

3.2 Mendelian Independent Inheritance in the Ed PGI

Moving from a single variant to the PGI simply means summing across the same random-segregation for all the variants. The Ed PGI is constructed by summing the 0, 1, or 2 copy indicators across the $J \approx 8,000$ different SNPs, each multiplied by the externally estimated weights $w_{j'}$ that capture how strongly each SNP predicts education years. Because every variant copy enters the child's genome through an independent random draw (conditional on parental values), the child's Ed PGI is the weighted sum of thousands of such draws.

This means the Ed PGI can be decomposed into the component that was inherited as a result of a random draws from parent(s) who had 1 copy of the variant, and the certainty

[7]Note both parents having 0 or 2 copies leaves $x_{i',j'}^{\text{Random}} = 0$, so that individual i' inherited no random variation in variant j'.

component from draws where parent(s) had 0 or 2 copies.

$$Z_i^{\text{Child}} = Z_i^{\text{Random}} + \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}\right],$$

where $Z_i^{\text{Random}} = \sum_{j=1}^{J} w_j x_{i,j}^{\text{Random}}$ is the random component of the Ed PGI for individual i, and similarly for their parents' Ed PGI values, $Z_{m(i)}^{\text{Mother}}$ and $Z_{f(i)}^{\text{Father}}$. Z_i^{Random} is the random component because it was the result of Mendelian independent inheritance, and its only determining factor was having at least one parent with exactly one copy for relevant genetic variants — and nothing else.

Ideally, the random genetic component Z_i^{Random} is constructed from observed parental Ed PGI values, $Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}$. However, parental PGI values are not observed for a large enough sample in the UKB. Thus, I follow Young et al. (2022) and leverage siblings' genetic data to impute parental PGI values. This imputation recovers about 75% of the parental genetic information, yielding imputed parental PGIs that are extremely highly correlated with the true parental values — see Subsection 2.3.

Formally, denote the Ed PGI value for individual i's sibling $s(i)$ as $Z_{s(i)}^{\text{Sibling}}$. The observed random component, using sibling PGI values (instead of parents') is defined as follows,

$$\widetilde{Z}_i^{\text{Random}} = Z_i^{\text{Child}} - \frac{1}{2}\left(Z_i^{\text{Child}} + Z_{s(i)}^{\text{Sibling}}\right) = \frac{1}{2}\left(Z_i^{\text{Child}} - Z_{s(i)}^{\text{Sibling}}\right).$$

The imputed random component is equivalent to using the variation between an individual and their observed sibling. This random component, therefore, is using the same within-family variation that early genetic studies did by examining within family differences. An individual's Ed PGI is drawn from the same distribution as their sibling, because they had the same parents; in absence of observing both parents directly, this approach uses the sibling to infer this distribution.

This approach is equivalent, in expectation, to using observed parents because of the simple fact that children and parents are two independent draws from the parental genetic distribution. Thus, in expectation, they both proxy for the parental PGI values,

$$\mathbb{E}\left[\frac{1}{2}\left(Z_i^{\text{Child}} + Z_{s(i)}^{\text{Sibling}}\right)\right] = \frac{1}{2}\left(Z_{m(i)}^{\text{Mother}} + Z_{f(i)}^{\text{Father}}\right) \Rightarrow \mathbb{E}\left[\widetilde{Z}_i^{\text{Random}}\right] = Z_i^{\text{Random}}.$$

By construction, $\widetilde{Z}_i^{\text{Random}}$ captures Mendelian independent variation, conditional on the sibling's observed genetic values. However, because siblings collectively reveal approximately

75% of the parental genetic variation, the resulting imputed random component $\widetilde{Z}_i^{\text{Random}}$ contains measurement error compared to the ideal (but unobserved) Z_i^{Random}.

The sibling-imputed parental mean PGI can be viewed as a proxy for the true parental mean PGI,

$$\frac{1}{2}\left(Z_i^{\text{Child}} + Z_{s(i)}^{\text{Sibling}}\right) = \frac{1}{2}\left(Z_{m(i)}^{\text{Mother}} + Z_{f(i)}^{\text{Father}}\right) + \nu_i,$$

where ν_i denotes the measurement error arising from the incomplete observation of parental genotypes. This error, ν_i, comes from a sibling random inheriting more or fewer genes from one parent than the other, and is an independent draw from the primary individual by construction. Notably, however, this error reduced precision of using the imputed proxy measure, compared true parental values

Consequently, the derived random genetic component $\widetilde{Z}_i^{\text{Random}}$ maintains the same core identification properties as the ideal Z_i^{Random} — notably, Mendelian independent inheritance — though it inherently includes measurement error due to sibling-based imputation. Its precision therefore depends on how genetically distinct each sibling pair is: sibling pairs who differ widely provide clearer signals, whereas genetically similar pairs introduce greater measurement noise. Using $\widetilde{Z}_i^{\text{Random}}$ as an IV yields consistent, but less precise estimates compared to the ideal scenario of directly observed parental genotypes. For estimation, an Ordinary Least Squares (OLS) regression will implicitly weight observations by variance (Angrist 1998), meaning sibling pairs with larger genetic differences disproportionately influence the estimated effects. Thus, while the sibling-based imputation approach preserves unbiasedness in estimating genetic effects, it comes at the cost of increased standard errors and potentially reduced generalisability if sibling genetic divergence is not representative for the entire population.

3.3 Measurement Error in the Ed PGI

Previous studies have treated observed Ed PGI values as if they were perfect genetic measures, not accounting for the fact that $Z_i^{\text{Child}} = \sum_{j'=1}^{J} w_{j'} z_{i,j'}$ is calculated from many statistical estimates of $w_{j'}$, which are themselves subject to measurement error. Although the Okbay et al. (2022) weights are estimates from an exceptionally large sample of over 3 million individuals, they remain subject to finite sample sampling variation and estimation error; the sample size is large, but the number of estimated parameters is similarly large ($w_{j'}$ for around 8,000 different values), making estimation error a sufficient concern.

Correspondingly, any statistical analysis using observed values of Z_i^{Child}, or $\widetilde{Z}_i^{\text{Random}}$, will

suffer from attenuation bias and overconfident standard errors, which do not account for the statistical uncertainty in the GWAS weights $w_{j'}$. This even holds true in an IV specification, because Z_i^{Child} and $\widetilde{Z}_i^{\text{Random}}$ are measured with the same measurement error, because they are both functions of the GWAS weights $w_{j'}$. In this case, the IV and endogenous variable have correlated measurement error, leading to attenuation bias in IV estimates.

To avoid this form of bias, I calculate $\widetilde{Z}_i^{\text{Random}}$ using a set of GWAS weights estimated in a different population to that for Z_i^{Child}. Specifically, I take a secondary set of weights also provided in the supplementary data for Okbay et al. (2022), that were estimated exclusively from the UK Biobank subsample (and not the 23&Me sample).[8] This means uses an imputed random component which has measurement error coming from a different population than that for the Ed PGI, thus purging the attenuation bias in IV estimates.

This approach was first proposed in the genetics literature, to purge bias from PGI correlations (van Kippersluis et al. 2023) and is applied here for the first time for causal estimates. Using a PGI with one set of GWAS weights as an IV for another PGI constructed with different weights raises a key question: which set of weights should define the random (instrumental) component, and which should be used for the endogenous Ed PGI? The method of Obviously Related Instrumental Variables (ORIV) removes this subjective choice by estimating the IV specification twice, once with each set of weights serving as the instrument for the other set (Gillen et al. 2019). ORIV then combines the estimates optimally and clustering the standard errors to account for the two-step estimation (i.e., clustering on individuals). This yields unbiased estimates while efficiently accounting for estimation uncertainty from both sets of GWAS weights — see van Kippersluis et al. 2023 for full details.

3.4 Causal Identification Strategy

Formally, let $D_i(z')$ denote the potential education level and $Y_i(z', D_i(z'))$ labour market outcome for individual i, under the hypothetical alternative realisation that i had Ed PGI value $Z_i^{\text{Child}} = z'$. Observed education and wages/income data are a sum of their respective potential outcomes.

$$D_i = \sum_{z'} \mathbb{1}\left\{Z_i^{\text{Child}} = z'\right\} D_i(z'), \quad Y_i = \sum_{z'} \mathbb{1}\left\{Z_i^{\text{Child}} = z'\right\} Y_i(z', D_i(z')).$$

[8] Note that this analysis also uses the UKB data, so that this secondary Ed PGI is calculated on a sample that overlaps the analysis sample. This will lead to over-fitting, although is ameliorated by the fact that the analysis sample is the sibling sample, about 10% of the entire UKB, limiting the overlap to the extent possible.

Note that $Y_i(z', D_i(z'))$ is a function of both hypothetical Ed PGI value, and resulting education $D_i(z')$; Section 5 returns to this distinction.

Target Parameters. This study aims to estimate the average causal effect of having a higher Ed PGI, which is summarised by the following average partial effects:

$$\frac{\partial}{\partial z'}\mathbb{E}\left[D_i(z')\right], \quad \frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(z', D_i(z'))\right].$$

The first average partial effect represents the average effect of a one standard deviation change in the Ed PGI on a measure of education, and the second on labour market wages or income.

The Ed PGI, however, is neither randomly assigned nor directly manipulatable, so that we may not observe $Y_i(z', D_i(z'))$ for different values of z' for the same individual, and causal effects are not identifiable without further assumptions (i.e., the fundamental problem of causal inference). Thus, I rely on random variation coming from the random component of the Ed PGI to infer causal effects.

Identifying Assumptions. I use the random component of the Ed PGI, $\widetilde{Z}_i^{\text{Random}}$, as an instrument for the Ed PGI, Z_i^{Child}. This requires a number of assumptions to recover valid average causal effects.

(1) Instrument Relevance and Independence: The random component of the Ed PGI strongly predicts the individual's Ed PGI, and is unrelated to potential outcomes.

(2) Instrument Monotonicity: A higher value of the random component (weakly) increases the Ed PGI, with probability 1.

(3) Exclusion Restriction: The random genetic component, $\widetilde{Z}_i^{\text{Random}}$, affects the potential outcomes only via the Ed PGI, Z_i^{Child}.

The three identifying assumptions are justified by the biology of genetic inheritance and the design of the empirical strategy. The relevance and independence assumption (1) holds because the random genetic component arises solely from Mendelian inheritance, ensuring both predictive strength (see Table 3) and independence from unobserved confounding conditional on parental genetics and observed covariates. The monotonicity assumption (2) is satisfied because inheriting a higher random genetic component can only additively increase the Ed PGI, never reducing it. Indeed, the random component mechanically changes the Ed PGI for anyone who receives a higher random component, so that everyone is a complier, too. Finally, the exclusion restriction (3) is credible given that genetic variants used in the

random component influence socioeconomic outcomes only through their aggregation into the Ed PGI, by construction.

Estimation Strategy. I estimate the genetic first-stage as follows,

$$Z_i^{\text{Child}} = \gamma + \theta \widetilde{Z}_i^{\text{Random}} + \boldsymbol{\zeta}' \boldsymbol{X}_i + \eta_i, \tag{2}$$

where θ is the first-stage average contribution of the random component (with measurement error), and $\boldsymbol{X}_i$ is a vector of controls including parental imputed PGI levels, gender, age, sibling count, in city indicator, and the other PGIs (i.e., all demographic variables in Figure A1). Since $\widetilde{Z}_i^{\text{Random}}$ serves as a formula instrument, it must be re-centred with respect to the control variables to produce valid causal effects (Borusyak & Hull 2023). I do this by calculating $\mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_{m(i)}^{\text{Mother}}, Z_{f(i)}^{\text{Father}}\right]$ with interactions for the imputed parental PGI values and demographic variables.

I estimate the second-stage with a regression specification as follows, where $\boldsymbol{X}_i$ is a vector of control variables described previously.

$$Y_i = \alpha + \beta Z_i^{\text{Child}} + \boldsymbol{\varphi}' \boldsymbol{X}_i + \varepsilon_i. \tag{3}$$

β represents the average effect of the Ed PGI on the education variables, and on wages and annual income (measured in log values). I use the Ed PGI random component, $\widetilde{Z}_i^{\text{Random}}$, to instrument for the Ed PGI in the IV specifications, using the ORIV method (with Ed PGI calculated using two independent weights) where mentioned. The IV first-stage is shown in Equation (2), with the same set of control variables as the second-stage.

Limitations. The OLS and IV estimates place greater weight on individuals with larger deviations in the random component, $\widetilde{Z}_i^{\text{Random}}$, which might not represent effects for the entire population — potentially limiting external validity. Additionally, despite independent Mendelian inheritance, the random genetic component remains modestly correlated with other PGIs. While I control for these other PGI values, these correlations further constrain generalisability beyond the analysis sample, by implying overlap with the Ed PGI and these other PGIs. Finally, the linear specifications employed (OLS and ORIV) yield weighted averages of individual-level causal effects. This linear specification might not fully reflect heterogeneity in causal effects across the PGI distribution, or provide weights for the average causal effects representative for the entire UKB population — see Blandhol et al. (2022) for further details.

4 Genetic Effects

The Ed PGI, Z_i^{Child}, is not randomly assigned — individuals inherit genetic variants from their parents, creating systematic correlations with family background. However, by isolating the random variation in genetic inheritance I recovers credible estimates of the average causal effects of the Ed PGI on education and labour market outcomes. This section shows the empirical estimates of genetic effects.

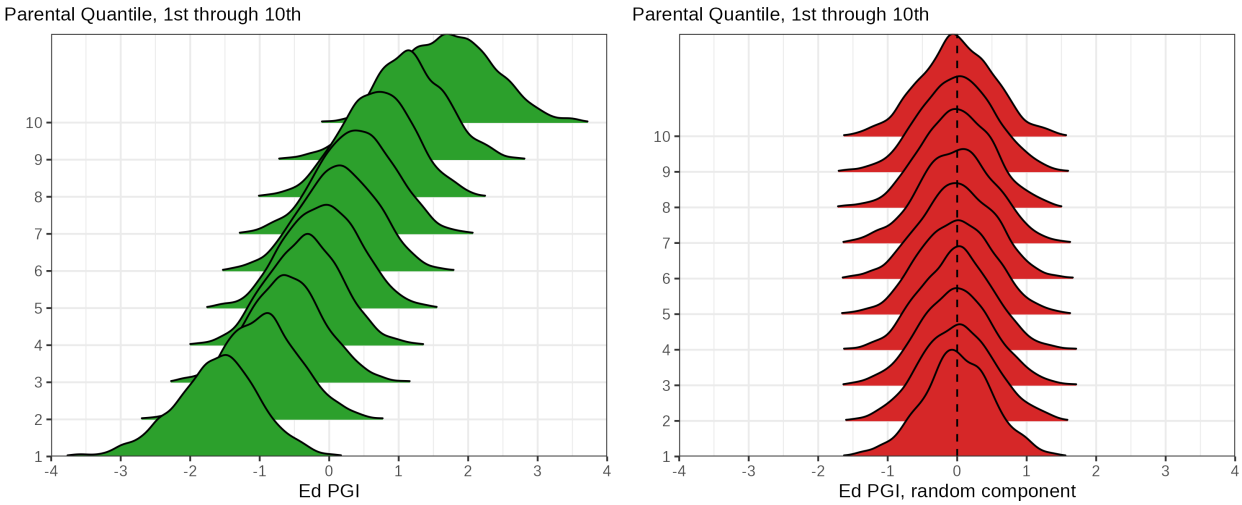
4.1 Genetic First-Stage Results

The Ed PGI imputed random component, $\widetilde{Z}_i^{\text{Random}}$, serves as a formula instrument for an individuals' Ed PGI, Z_i^{Child}.

Describe the random component. What is its distribution. It is not correlated with other demographic variables, and less so with the other PGIs. Figure 4 shows the distribution, and Figure A1 shows what it is correlated with.

Figure 4: Distribution of the Ed PGI and the Random Component, Relative to Imputed Parental Decile.

(a) Ed PGI, Z_i^{Child}. **(b)** Ed PGI random component, $\widetilde{Z}_i^{\text{Random}}$.



Note: This figure shows the distribution of Ed PGI and the random component, Z_i^{Child} and $\widetilde{Z}_i^{\text{Random}}$, conditional on the decile of the parental imputed Ed PGI. Parental imputed Ed PGI decile 1 shows the distributions for the bottom 10% of parental imputed values, decile 2 shows distribution among 10–20%, and so on.

The Ed PGI random component is a strong instrument for the Ed PGI, even when

using the primary as an instrument for the secondary (1 → 2, and visa versa). Table 3 summarises estimates of the genetic first-stage defined in equation (2). Columns (1) and (5) show mechanical results, equal to one by construction, as the random component is calculated from the primary and secondary Ed PGIs respectively. Columns (2) and (4) show that the random component is a strong instrument for the Ed PGI, calculated using weights from the other sample (primary or secondary), confirming that the random component from one PGI retains predictive power for the other, independently calculated PGIs. The fact that these cross-PGI coefficients are not equal to one indicates substantial measurement error in both PGIs, which would lead to attenuation bias in standard IV estimation. The ORIV estimates, columns (3) and (6) explicitly addresses this measurement error by combining information from both primary and secondary first-stages. Although ORIV estimates recover unbiased coefficients, this approach necessarily comes at the expense of greater variance.

Table 3: First-stage Genetic Estimates, Random Component on Ed PGI ($\widetilde{Z}_i^{\text{Random}} \to Z_i^{\text{Child}}$).

	Outcome:					
	Primary Ed PGI, Z_i^{Child}			Secondary Ed PGI, Z_i^{Child}		
	(1)	(2)	(3)	(4)	(5)	(6)
Primary Ed PGI, Z_i^{Random}	1			0.204		
	(0.001)			(0.012)		
Secondary Ed PGI, Z_i^{Random}		0.158			1	
		(0.011)			(0.001)	
Combined, ORIV			1.336			1.495
			(0.043)			(0.048)
F-statistics		207	971	305		957
R^2	0.998	0.091		0.051	0.994	
Observations	24755	24755	24755	24755	24755	24755

Note: Shows estimates of the genetic first-stage, (2), using the random component as an instrument for the Ed PGI. The first three columns using the primary Ed PGI as the outcome, while the next use the secondary Ed PGI. The row marked ORIV uses the combined estimates of the primary and secondary Ed PGI random component on the Ed PGI, estimated using ORIV.

Table 3 shows how much

Points raised by Prof Miller:

1. Also show $Z_i^{\text{Child}} \to Z_i^{\text{Child}}$, or random → random in appendix table, to get a better sense about what this implies.

2. What does the first-stage attenuation bias say about the reliability ratio of commonly used Ed PGI values? It says the reliability is fairly low....

4.2 Genetic Effect Results

The following results section begins by presenting ordinary least squares (OLS) estimates to document the correlational relationships between the Education PolyGenic Index (Ed PGI) and various education and labour market outcomes. Table 4 first displays raw correlations without any controls (Column 1). Column 2 adds an extensive set of demographic controls to these specifications, and Column 3 further corrects for attenuation bias arising from measurement error by implementing a stacked OLS specification that uses a secondary set of GWAS weights.

Table 4: OLS and IV Estimates, Effect of Ed PGI on Education, Occupation Wages and Income.

	OLS			IV			
	OLS	OLS + controls	OLS Stacked	Reduced Form	Regular IV	ORIV	Observation Count
Outcome	(1)	(2)	(3)	(4)	(5)	(6)	
Education years	1.1	1.1	1.5	0.10	0.60	0.80	24755
	(0.02)	(0.02)	(0.05)	(0.04)	(0.21)	(0.08)	
University degree	0.14	0.14	0.19	0.01	0.07	0.10	24755
	(0.00)	(0.00)	(0.01)	(0.01)	(0.03)	(0.01)	
Occupation hourly wage	0.09	0.09	0.12	0.01	0.04	0.07	24755
	(0.00)	(0.00)	(0.01)	(0.00)	(0.03)	(0.01)	
Occupation annual income	0.08	0.09	0.12	0.00	0.03	0.06	20602
	(0.00)	(0.00)	(0.01)	(0.01)	(0.04)	(0.02)	

Note: Shows estimates of the genetic effects, of (3). The first 3 columns show the OLS correlation between Ed PGI and various outcomes; columns (3) through (6) show point estimates from the IV model, using the random component as an instrument for the ED PGI. OLS stacked refers to using the obviously related method to estimate the correlation between Ed PGI and outcome, purging bias from measurement error; similarly for the column named ORIV, with the IV approach for causal effects. For example, 0.74 in row 1 column (6) means a 1 Standard Deviation (SD) increase in the Ed PGI (motivated by Mendelian independent assortment) leads to 0.74 extra years. The regressions for income variables are logged, so the point estimate means a percent increase in income for every 1 SD increase in the Ed PGI.

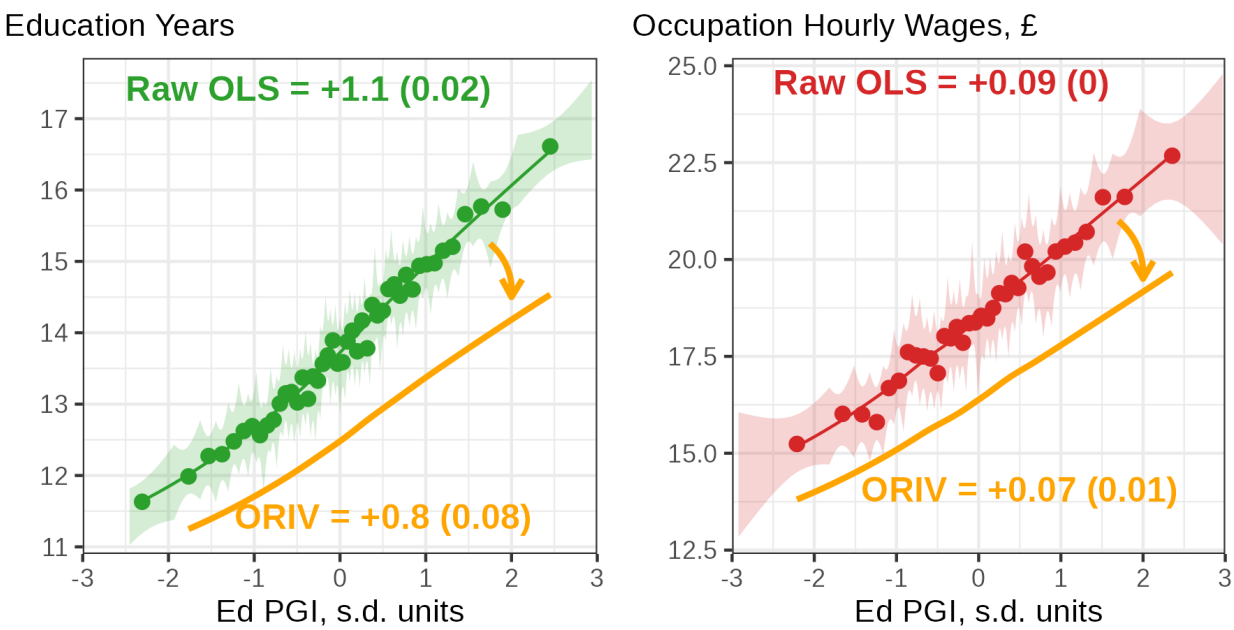
The addition of demographic controls makes negligible differences to the correlation estimates across outcomes, indicating minimal confounding by the observed demographic

factors. However, adjusting for measurement error significantly increases the correlation estimates. Specifically, correlations between the Ed PGI and education outcomes (education years and having a university degree) increase by approximately 30-40%, while those for labour market outcomes (hourly wage and annual income) increase by roughly 20-40%. These adjusted estimates substantially exceed previously reported correlations for similar variables in earlier studies, such as Papageorge & Thom (2020).

The IV estimates isolate causal effects by using the inferred Mendelian random component as an instrument for the Ed PGI. Notably, these IV estimates are systematically lower than their corresponding OLS correlations. This discrepancy implies that growing up in a family environment shaped by parents with higher Ed PGI values independently and positively influences children's educational attainment and labour market outcomes, beyond the direct genetic influences captured by the Ed PGI.

Figure 5: OLS and IV Estimates, Effect of Ed PGI on Education and Wages.

(a) Ed PGI → Education Years. **(b)** Ed PGI → Hourly Wages.



Note: This figure shows the correlation between Ed PGI and Education Years, as well as occupation coded Hourly Wages. The green (or red) coloured lines are the obviously related estimated correlation between the Ed PGI and the respective outcomes, and the yellow lines with a slope matching the ORIV causal effect estimates.

The IV estimates confirm meaningful average causal effects of the Ed PGI. For example, a one-standard-deviation increase in the Ed PGI causally leads to an additional 0.74 years of education and about a 6.1% increase in occupation coded hourly wages, according to the

ORIV estimates. Adjusting for measurement error notably affects these causal estimates, increasing them by roughly 30-40% when moving from standard IV (Column 4) to ORIV specifications (Column 6), validating the importance of addressing measurement error. These causal estimates are meaningfully larger than those recently presented by Carvalho (2025), who analysed the same data but did not account for measurement error.

Correcting for measurement error comes at a cost: the standard errors in ORIV specifications are noticeably larger than those in the unadjusted IV columns. This increase reflects proper acknowledgment of statistical uncertainty in the GWAS weights used to construct the Ed PGI. Prior studies reporting smaller standard errors do not account for this additional uncertainty, and thus substantially overstate the precision and reliability of their correlation or causal effect estimates.

4.3 Within Family Variation

Compare results with and without family FEs. Gives an idea that with families there is still a lot of genetic variation.

Describe that using other PGIs as controls (including the ADHD PGI) makes little difference. Cite that other paper which mentions controlling for other PGIs as if it is important.

Compare results for families with at least one parent. Measurement error from the sibling imputation is lower among these people (though not zero), so might imply that

5 Direct and Indirect Effects

The Ed PGI exhibits clear labour-market effects in the data. IV estimates that isolate the random component of genetic inheritance indicate that a one-standard-deviation increase in the Ed PGI causally raises education and improves labour earnings: the preferred ORIV specifications imply roughly 0.74 additional years of education and about a 6.1% increase in occupation-coded hourly wages per s.d. of the Ed PGI. These causal effects exceed earlier reports that did not correct for measurement error in PGI weights, consistent with the stacked/ORIV adjustments materially inflating both education and earnings relationships once attenuation is addressed.

The size and robustness of these labour-market effects has led to a common interpretation that at least part of the Ed PGI's influence must be direct, operating through traits such

Table 5: OLS and IV Estimates, Effect of Ed PGI on Education, Occupation Wages and Income, with Family Fixed Effects.

	OLS			IV			
	OLS	OLS + controls	OLS Stacked	Reduced Form	Regular IV	ORIV	Observation Count
Outcome	(1)	(2)	(3)	(4)	(5)	(6)	
Education years	0.55	0.56	0.79	0.11	0.66	0.79	24714
	(0.039)	(0.039)	(0.083)	(0.036)	(0.210)	(0.083)	
University degree	0.069	0.070	0.100	0.015	0.090	0.100	24714
	(0.0060)	(0.0061)	(0.0120)	(0.0056)	(0.0320)	(0.0120)	
Occupation hourly wage	0.0510	0.0510	0.0670	0.0067	0.0390	0.0660	24714
	(0.0056)	(0.0055)	(0.0110)	(0.0051)	(0.0300)	(0.0110)	
Occupation annual income	0.0510	0.0580	0.0780	0.0085	0.0490	0.0780	20570
	(0.0110)	(0.0100)	(0.0190)	(0.0095)	(0.0550)	(0.0190)	
Family Fixed Effects?	Yes	Yes	Yes	Yes	Yes	Yes	

Note: This table shows estimates of the genetic effects, Equation (3) with various different specifications — now, all including family fixed effects in the estimates. All columns are as in Table 4, and income variables are logged.

as intelligence rather than exclusively through education. This view is often motivated by the fact that the Ed PGI predicts fluid-intelligence scores and by results that show earnings associations even when conditioning on education (Carvalho 2025). Such claims appear, for example, in work that argues the Ed PGI captures skills related to financial literacy and other non-educational attributes (Barth et al. 2020).

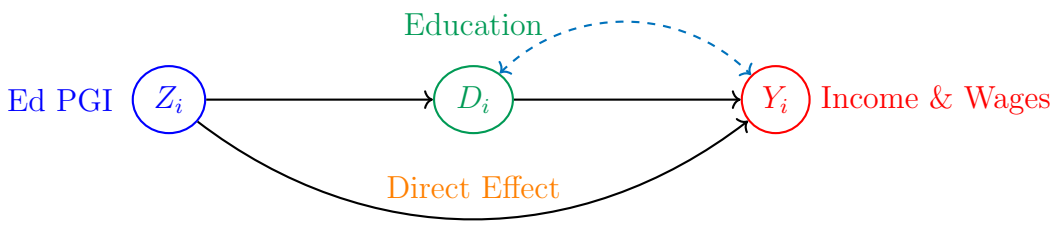
This section examines those claims with a Causal Mediation (CM) framework to separate direct genetic effects from indirect effects via take-up and returns to education. The direct and indirect average partial effects are defined as follows, where Y_i is again a measure of labour market wages or income, and D_i a measure of education (education years or a binary measure for an education qualification).

$$\text{Direct Effect } = \frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(z', D_i(Z_i^{\text{Child}}))\right], \quad \text{Indirect Effect } = \frac{\partial}{\partial z'}\mathbb{E}\left[Y_i(Z_i^{\text{Child}}, D_i(z'))\right].$$

The direct effect corresponds to the average partial effect of the Ed PGI on wages or income while holding the education channel D_i fixed; the indirect effect captures the average partial effect that operates through the education response $D_i(z')$. This decomposition maps

to the empirical question raised above: "do education-linked genes affect earnings directly, or mainly by shifting the education margin?"

Figure 6: Structural Causal Model for Direct and Indirect Effects.



Note: This figure shows the structural causal model behind direct and indirect effects, where $Z_i \to D_i$ represents a causal relationship of Z_i on D_i. The dotted line between education and income & wages represents the fact that education is not randomly assigned, and a strategy for identifying returns to education is needed for valid CM estimates.

Valid inference on CM effects requires valid causal estimates of returns to education, to avoid selection bias in CM estimates and contaminated inference (see Hogan-Hennessy 2025).

Education is not randomly assigned, so I proceed on two complementary avenues for credible returns to education estimates: (i) exploiting the 1972 British Minimum School-Leaving Age (MSLA) reform, which induces an extra year of schooling to estimate education returns for MLSA compliers; and (ii) sibling fixed-effects estimates, which offer conservative, design-consistent benchmarks within the UKB sibling analysis sample. These ingredients are used below to interpret the mediated component of the Ed PGI's labour-market effect, and to assess the scope for a direct effect.

Also note that one cannot use the ORIV to purge measurement error in this analysis, as it makes everything too imprecise for meaningful inference. So this part is using estimates that suffer from measurement error; if one assumes that average direct and indirect effects suffer equally from attenuation bias, then the proportion going through the indirect education channel is representative.

5.1 Minimum School Leaving Age (MSLA)

In 1972, the UK increased the Minimum School Leaving Age (MSLA) by one year, from age 15 to 16. The change took effect at the start of the 1972-73 school year, and applied to those who were 15 or younger on 1 September 1972, i.e., those born on or after 1 September 1957. The reform therefore induced extra school for individuals born around date 1 September

1957. At the end of the school year which became compulsory, British school children sit exams for the GCSE qualification, so the reform pushed a larger share of the UKB cohorts to remain in school to 16 and to complete these qualifications. The 1972 MLSA change has been previously used for labour market returns to education (Oreopoulos 2006) and health returns to education (Clark & Royer 2013). The UKB population overlaps well with the birth date cut-off, and so has been used in a series of papers for estimating returns to education in the UKB (Barcellos et al. 2021, 2023).

The analysis treats exact birth month as the running variable in a fuzzy regression discontinuity design (RDD), using the indicator for being born September 1957 and after as an instrument. Because not all those eligible necessarily stayed beyond 15, the post-cutoff indicator instruments (i) GCSE completion, in specifications focussed on credentials, (ii) education years, where mentioned. I estimate the following RDD equation, to estimate the effect of the MSLA change on education,

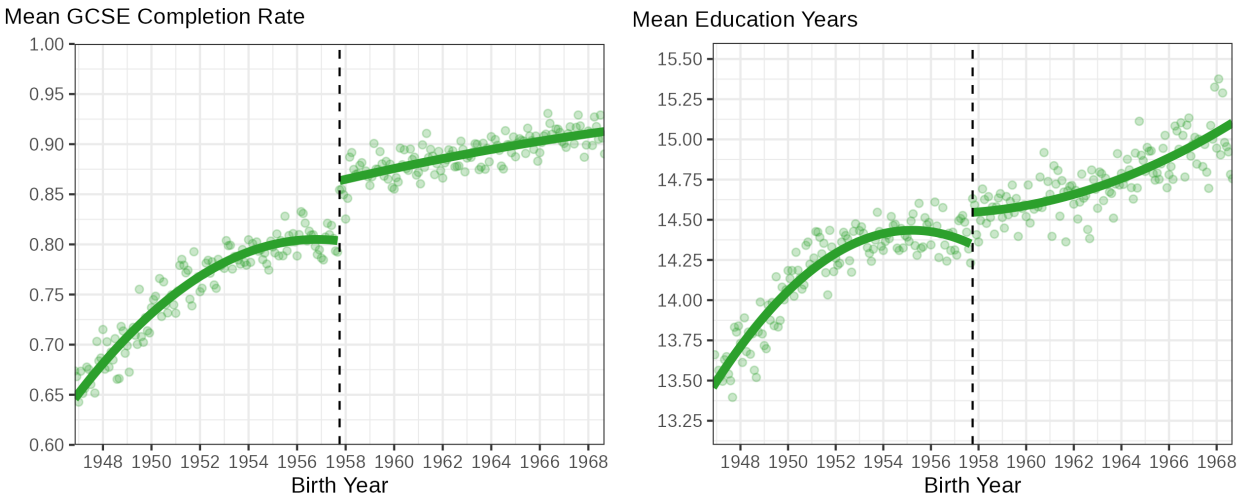
$$D_i = l_0(b_i) + \mathbb{1}\left\{1957.75 \leq b_i\right\} + l_1(b_i) + \epsilon_i$$

where b_i represents individual i's birth year plus birth month divided by 12 (so that 1957.75 represents born in September 1957), and l_0, l_1 are control trends either side of the cut-off.

I estimate the RDD equation with triangular kernel weights (weighting closer to the cut-off), and in two main specifications — one with a linear trend before and after the cutoff and a bandwidth of 5 years, and another with a polynomial in birth month, and a bandwidth of 10 years (following Barcellos et al. 2021). The polynomial specification thus has more observations, as it includes people who are born further before or after the MSLA change.

A practical constraint concerns sample size around the 1 September 1957 threshold. The sibling subsample is too small to deliver precise RD estimates for years of education: there are too few sibling observations in the immediate neighbourhood of the cutoff to identify a stable first-stage effect on education. For the RD analysis, the estimation therefore uses the broader UK Biobank cohort — restricting to births within 5 or 10 years of the cutoff — to increase the density of observations close to the threshold while preserving local comparability. This design yields substantially more information at the policy margin and improves power for both the discontinuity in schooling and the corresponding fuzzy RD/IV estimates. All RD results are reported with both specifications, showing that conclusions are unchanged across the 5 and 10 year windows; SEs are clustered two-way at the month + birth year level. Figure 7 shows the effect of the MSLA change on GCSE completion rate and mean education years, using a squared polynomial either side of the cut-off and a bandwidth of 10

Figure 7: Mean Education Completion Around the MSLA Rise.



Note: These figures show the first-stage education effects of the 1972 MSLA rise, where individuals born after September 1957 (the dashed line) were leggally compelled to stay in school until 16, whereas before only until age 15. Birth cohort trends were calculated with a squared polynomial before and after the cutoff. These estimates are calculated from all UKB respondents, born within 10 years of the cutoff and who have reported years of education.

years before and after September 1975.

The policy generated a clear discontinuity in schooling at the date-of-birth cutoff: the share completing the GCSE qualifications rose after for students born in September 1957 and afterwards, with a corresponding (smaller) increase in mean education years These first-stage effects are visible in UK Biobank and are consistent with nationally representative evidence on the same reform — and previous research that estimate this same cut-off.

Panel A of Table 6 documents clear discontinuities at the September 1957 cutoff. In the narrower bandwidth with linear cohort trends (columns 1 and 3), the post-cutoff indicator increases the probability of completing GCSE/O-level by 5.3 p.p. (s.e. 0.2 p.p.) and raises education years by 0.122 (s.e. 0.016) years. In the wider bandwidth with quadratic trends (columns 2 and 4), the corresponding effects are 6.0 p.p. (0.7 p.p.) and 0.201 (0.033) years. Reduced-form impacts on log hourly wages and log annual income are mechanically small and positive—0.006–0.011 and 0.001–0.014 log points, respectively—consistent with modest average shifts in schooling. IV relevance is very strong for the education outcomes (e.g., F = 1049, 79 for GCSE completion; 56, 37 for education years).

Using the post-cutoff indicator as an instrument yields estimates for labour market returns to education among MLSA compliers on the leaving-age margin, shown in Panel B of Table 6. For log hourly wages, OLS estimates imply 6.1% per additional year of education across both

Table 6: First-stage of the MSLA Change, and Returns to Education IV Estimates.

Panel A: First-stage.								
	First-stage.				Reduced form.			
Outcome:	GCSE Completion		Education Years		Hourly Wage		Annual Income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
MSLA Effect	0.053	0.060	0.122	0.201	0.006	0.011	0.001	0.014
	(0.002)	(0.007)	(0.016)	(0.033)	(0.002)	(0.003)	(0.008)	(0.003)
F-statistics	1049	79	56	37	5	16	0	26
Panel B: Second-stage.								
	OLS.				MLSA IV.			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Outcome: Log Occupation Hourly Wage								
GCSE completion	0.258	0.260			0.104	0.183		
	(0.009)	(0.009)			(0.043)	(0.028)		
Education years			0.061	0.061			0.045	0.055
			(0.000)	(0.001)			(0.008)	(0.007)
Outcome: Log Occupation Annual Income								
GCSE completion	0.272	0.262			0.027	0.231		
	(0.010)	(0.009)			(0.086)	(0.042)		
Education years			0.066	0.065			0.012	0.071
			(0.001)	(0.001)			(0.037)	(0.008)
Linear trends	Yes		Yes		Yes		Yes	
Polynomial trends		Yes		Yes		Yes		Yes
Observations	142,895	245,857	142,895	245,857	142,895	245,857	142,895	245,857

Note: Talk about it. Note this is among the entire sample, and not the sibling analysis subsample.

bandwidths, and the MLSA IV estimates lie in the 4.5–5.5% range. For log annual income, OLS returns are 6.5–6.6%, while the IV range is 1.2–7.1% across specifications. For labour market returns to to GCSE completion rate, OLS associations are 0.258–0.272 log points and the IV estimates 0.104–0.231. Overall, the IV returns are of similar order of magnitude to their OLS counterparts, supporting the use of these MLSA-anchored LATEs as credible benchmarks for the mediation exercise.

5.2 OLS and Sibling Fixed Effects

Write a section how correlational estimates give an estimate for returns to education, which are often not so different from causally identified version.

And then estimates which control for family variation by fixed effects (siblings FEs) give lower bounds for returns to education.

Table 7: Returns to Education, OLS and Sibling FE Estimates.

	OLS				OLS with Sibling FEs			
Outcome:	Wage		Income.		Wage		Income.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Education years	0.062		0.062		0.046		0.049	
	(0.001)		(0.002)		(0.002)		(0.003)	
University degree		0.335		0.372		0.257		0.307
		(0.006)		(0.011)		(0.012)		(0.026)
R^2	0.255	0.121	0.182	0.099	0.739	0.730	0.767	0.764
Observations	24,755	20,602	17,894	15,046	24,755	20,602	17,894	15,046

Note: Talk about it. Note this is among the sibling analysis subsample.

Together, these strategies give credible returns to education among the UKB sample, which purges selection bias from CM estimates in this observational setting.

6 Direct and Indirect Effect Estimates

Long section describing the mediation estimates.

Use estimate the mediation effects, using (1) OLS (2) sibling FEs (3) MSLA.

Plan one further figure to illustrate the mediation estimates. Do so with the bar chart from previous code, in HRS data: OLS, sibling FE, IV.

7 Summary and Concluding Remarks

Conclusion goes here.

Table 8: Direct and Indirect Effect Estimates, with Multiple Levels of Education.

	OLS				OLS with Sibling FEs		MLSA IV	
Outcome:	Wage		Income.		Wage		Income.		Wage	Income.
	(1)	(2)	(3)	(4)	(5)	(6)				
Direct, Ed PGI effect										
GCSEs channel										
University degree channel										
Indirect, education effect										

Note: Talk about it. Note this is among the sibling analysis subsample.

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A Supplementary Appendix

This project used computational tools which are fully open-source. Any comments or suggestions may be sent to me at seh325@cornell.edu, or raised as an issue on the Github project.

A number of statistical packages, for the R language (R Core Team 2023), made the empirical analysis for this paper possible.

- *Tidyverse* (Wickham et al. 2019) collected tools for data analysis in the R language.

A.1 Occupational Imputed Wages and Income

Describe here the Kweon Koellinger imputation procedure. Reference their supplementary data, and thanks for additional guidance on it.

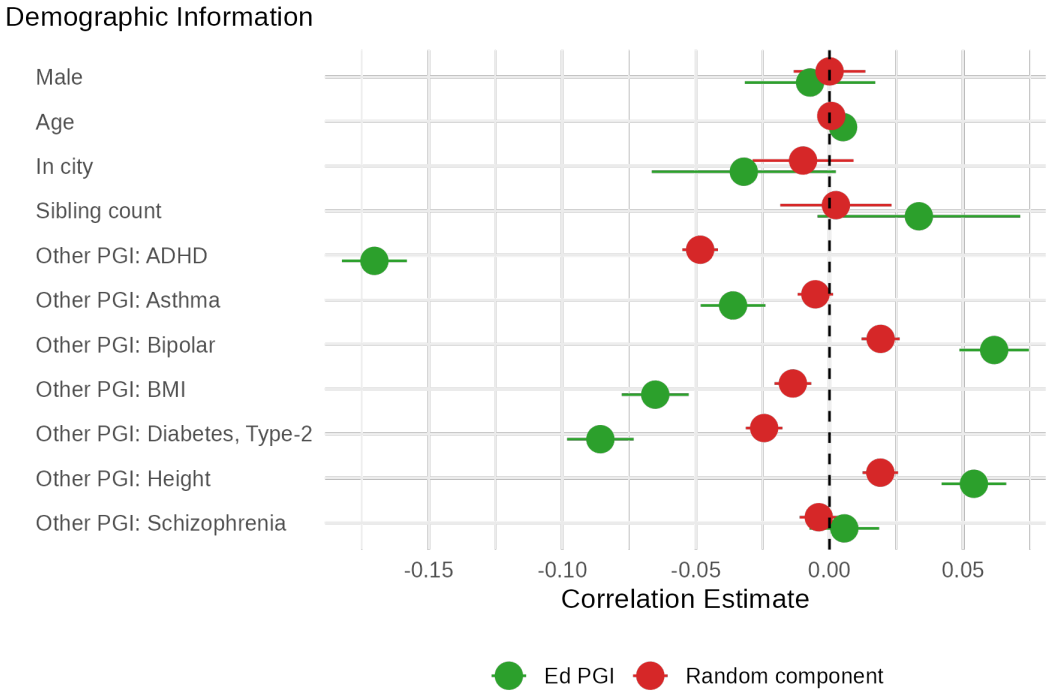
A.2 Identifying Family Clusters by Genetic Similarity

Describe here.

A.3 Validation of the parental PGI Imputation

A.4 Two PGIs as Obviously Related Instrument Variables

Figure A1: Correlation for Ed PGI, or Random Component, and Demographic Variables.



Note: Multivariate regression estimates for the linear model $Z_i = \boldsymbol{\phi}' \boldsymbol{X}_i + \eta_i$, for each Z_i^{Child} and $\widetilde{Z}_i^{\text{Random}}$, where $\boldsymbol{X}_i$ is the vector demographic variables described by the labels on the y-axis.